LINEAR TRANSFORMATIONS

Linear or not

Both conditions must hold for every choice of vectors and scalars.

LINEARITY

$$T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v}) \text{ and } T(c\mathbf{v}) = cT(\mathbf{v})$$

HOLDS WHEN for all $\mathbf{u}, \mathbf{v} \in V$ and all scalars c

✓ Linear

$(a, b) \mapsto (a, -b)$ with real scalars

Additive, and keeps every real multiple: the reflection of the plane is linear.

BOUNDARY

✗ Not linear

$T(z) = \bar{z}$ on $\mathbb{C}$ with complex scalars

Additive, but $T(i \cdot 1) = -i$ while $i\,T(1) = i$: homogeneity fails for $c = i$.

A map checked on a few particular vectors only

Verifying linearity on specific examples does not suffice; disproving it needs only one counterexample.

WHAT MOVES THE LINE

scalars Which numbers count as scalars is part of the definition. Additivity reads the same in both settings; homogeneity must be checked for the larger set.